

# Differential Calculus

## \*Home work sheet # 2

### Continuity and Differentiability

(a) Test the continuity of the following functions:

$$1. f(x) = \begin{cases} \cos x, & x \geq 0 \\ -\cos x, & x < 0 \end{cases} \quad \text{at } x = 0.$$

$$2. f(x) = \begin{cases} x \cos(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases} \quad \text{at } x = 0.$$

$$3. f(x) = \begin{cases} e^{1/x}, & x \neq 0 \\ 1, & x = 0 \end{cases} \quad \text{at } x = 0.$$

$$4. f(x) = \begin{cases} e^{\frac{-|x|}{2}}, & -1 < x < 0 \\ x^2, & 0 \leq x < 2 \end{cases} \quad \text{at } x = 0.$$

$$5. f(x) = \begin{cases} (x-a) \sin\left(\frac{1}{x-a}\right), & x \neq a \\ 0, & x = a \end{cases} \quad \text{at } x = a.$$

$$6. f(x) = \begin{cases} 1, & x < 0 \\ 1 + \sin x, & 0 \leq x < \pi/2 \\ 2 + (x - \pi/2)^2, & x \geq \pi/2 \end{cases} \quad \text{at } x = 0 \text{ and } x = \pi/2.$$

$$7. f(x) = \begin{cases} x \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases} \quad \text{at } x = 0.$$

$$8. f(x) = \begin{cases} \frac{|x-3|}{x-3}, & x \neq 3 \\ 0, & x = 3 \end{cases} \quad \text{at } x = 3.$$

$$9. f(x) = |x| + |x-1| \quad \text{at } x = 0 \text{ and } x = 1.$$

$$10. f(x) = \begin{cases} (1+x)^{1/x}, & x \neq 0 \\ 1, & x = 0 \end{cases} \quad \text{at } x = 0.$$

(b) Test the differentiability of the following functions:

$$1. f(x) = \begin{cases} x^2 \cos(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases} \quad \text{at } x = 0.$$

$$3. f(x) = |x| \quad \text{at } x = 0.$$

$$2. f(x) = \begin{cases} x^2 \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases} \quad \text{at } x = 0.$$

$$(c) \text{ Let } f(x) = \begin{cases} x^2 - 16x, & x < 9 \\ 12\sqrt{x}, & x \geq 9 \end{cases}.$$

Is  $f(x)$  continuous at  $x = 9$ ? Determine whether  $f(x)$  is differentiable at  $x = 9$ .

$$(d) \text{ Let } f(x) = \begin{cases} x^2, & x \leq 1 \\ \sqrt{x}, & x > 1 \end{cases}.$$

Is  $f(x)$  continuous at  $x = 1$ ? Determine whether  $f(x)$  is differentiable at  $x = 1$ .

$$(e) \text{ Show that } f(x) = \begin{cases} x^2 + 1, & x \leq 1 \\ x, & x > 1 \end{cases} \text{ is neither continuous nor differentiable at } x = 1. \text{ Also sketch the graph of } f(x).$$

*\*These problems are for the students only as home work. Search the reference books for more examples.*